

GROWRIDER · DESIGN DECISION RECORD

GrowRider TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	GRR-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/growrider		
Source	docs/decisions/0001-trl2-review-decisions.md @ d0c0d47		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D6; items O1 to O4 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-24) listed seven groups of items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open. The same instruction approved portfolio-wide decisions on the SwapCell interface, which do not apply to GrowRider (it has no battery), and confirmed that community designs pick co-design partners per area later.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-24) and in GRR-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Wheel size	20 in (ISO 406) for the whole fit range, rather than 24 in (ISO 507), which fails standover for 1.10 m riders. Decided by Amish, 2026-09-25: go with recommendation.
D2	Brakes	Coaster brake plus a front rim brake with a short-reach, reach-adjustable lever, rather than coaster only or two hand brakes. Decided by Amish, 2026-09-25: go with recommendation.
D3	Frame	Step-through steel frame with twin down tubes, rather than a small diamond frame. Decided by Amish, 2026-09-25: go with recommendation.
D4	Tires	Solid or airless puncture-proof tires for the prototype, rather than thorn-resistant tubes with liners; users to be asked in co-design. Decided by Amish, 2026-09-25: go with recommendation.
D5	Crank length	One 140 mm crank length for the prototype, with a possible 127 mm swap for the smallest riders later. Decided by Amish, 2026-09-25: go with recommendation.
D6	Production cost target	\$120 or less per bike at volume (R11), as proposed. Decided by Amish, 2026-09-25: go with recommendation. The prototype budget stays at \$250 in project.yaml .

Table 2. Items that remain open (no recommendation was made).

#	Item	Status
O1	Helmet funding (BOM line 20, \$12): fund it from the \$250 prototype budget (\$5 over), fund it separately, or raise the budget to about \$275	Proposed, awaiting Amish
O2	First partner: a bicycle distribution NGO, a rural school network or a regional bicycle assembler	Proposed, awaiting Amish (to be picked per area later, as Amish directed for community designs)
O3	First region: Zambia, Kenya or Malawi were suggested by the Buffalo parts ecosystem, with no single recommendation	Proposed, awaiting Amish
O4	Co-design partner for fit measurements and the regional parts list	Open; community designs pick co-design partners per area later

■ Consequences

- GRR-PRC-001 and GRR-REQ-001 are revised to v0.3: the four design choices are no longer "proposed", R4 names the 20 in wheel, and R11 carries the \$120 production target as decided. No requirement target was relaxed or redefined by these decisions, and `budget_usd` stays at \$250.
- The parametric model (`cad/src/model.py`), drawing GRR-DWG-001 and calculation note GRR-CAL-001 use the decided configuration.
- With the helmet still open, the prototype total is \$243 for the bike and \$255 with the helmet; the helmet line stays in the BOM, flagged.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.